

## SINGLE-LAYER MICROFLUIDIC PRESSURE GAIN VIA OPTOFLUIDIC LITHOGRAPHY

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In this work, pressure gain functionality in a single-layer microfluidic system is demonstrated for the first time. Autonomous microfluidic devices are essential for the long-term development of versatile biological and chemical platforms; however, creating effective control mechanisms has proven a significant challenge. For example, the signal degradation that results when multiple “Quake” valves are cascaded limits its usefulness in feedback devices [1], while adaptations that do allow for pressure gain suffer from time-consuming and laborious four-layer soft-lithographic fabrication processes [2]. Thus, here we introduce a single-layer microfluidic device – built from guided microstructures constructed *in situ* via optofluidic lithography – to enable a static pressure gain that is greater than unity. COMSOL simulations and analytical approaches were employed to characterize the effects of varying fundamental geometric properties on the pressure gain that can be achieved. Experimental results revealed pressure gains between  $1.94 \pm 0.09$  and  $6.30 \pm 0.23$ , for differential area ratios of 2 and 10, respectively. Such non-unity pressure gain could allow this technique to be used as a valve in microfluidic systems analogously to how transistors are used in electrical circuits.

**Figure 1a-c** shows conceptual illustrations of the single-layer microfluidic pressure gain system, which includes a free-moving piston that slides freely within a microchannel based on two primary fluidic terminals: (i) a high-pressure *source*, and (ii) a low-pressure *gate*. In the absence of an applied gate pressure, the piston rests in the “open” position (held in place by several ‘blocking’ microposts). When the gate pressure is increased beyond a critical threshold, the piston slides to the “closed” position. In addition, a third *exhaust* terminal serves to vent fluid from both the source and gate to atmospheric pressure, allowing the piston to move freely during this process. Because the piston includes two different cross-sectional areas – the *source area* ( $A_S$ ) and the *gate area* ( $A_G$ ) – a lower *gate pressure* ( $P_G$ ) can overcome a higher *source pressure* ( $P_S$ ) to slide the piston. This ability to achieve pressure gain is highly advantageous for microfluidic transistors.

**Figure 1d** illustrates the principle underlying the pressure gain mechanism. From the theoretical critical force required to close the piston (neglecting viscous drag and exhaust terminal effects), the

*critical gate pressure* ( $P_G^*$ ) can be approximated as:

$$P_G^* = \frac{A_S}{A_G} P_S = \frac{P_S}{R} \quad (1)$$

where  $R$  is the differential area ratio  $A_G/A_S$ . Consequently, the pressure gain can be calculated as:

$$\text{Pressure Gain} = \frac{P_S}{P_G^*} = \frac{A_G}{A_S} = R \quad (2)$$

For devices with a uniform piston height,  $R$  reduces to the ratio of the source and gate channel widths, thereby providing basic geometric parameters through which the pressure gain can be regulated as desired.

The microchannels were constructed via standard soft-lithography processes. Microchannels molded from poly(dimethylsiloxane) (PDMS) – with channel heights of  $60 \mu\text{m}$  – were bonded to glass slides coated with an  $80 \mu\text{m}$  layer of PDMS. **Figure 2** shows the completed pressure gain devices, and **Figure 3a-d** gives a conceptual illustration of the optofluidic lithographic process used to construct the gain piston structure within the bulk PDMS [5].

Fluid flow profiles were visualized using COMSOL Multiphysics laminar flow simulations. **Figure 4** depicts changes in pressure and fluid velocity around the piston. Although source-exhaust and gate-exhaust flows constitute minor fluid leakage, these flows can be reduced by decreasing the gap thicknesses relative to the piston size.

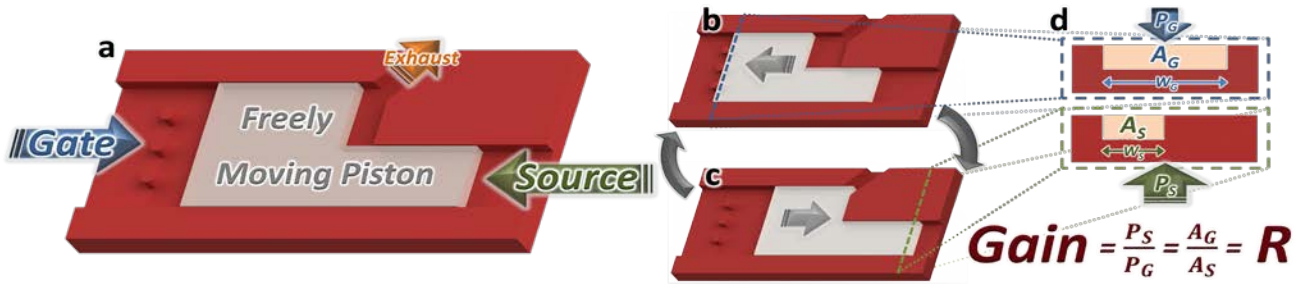
**Figure 5** shows experimental results for the threshold  $P_G^*$  required to close the piston for varying  $P_S$  (corresponding to devices with three distinct  $R$ 's). We observed maximum pressure gains for devices with  $R = 10$ , with measured gains of  $6.30 \pm 0.23$ .

Due to its ease of manufacture and relatively high gain, the presented single-layer pressure gain device could offer a simple, yet powerful component for a variety of microfluidic circuits and systems. For example, **Figure 6** shows a proposed design for a single-layer microfluidic pressure gain valve, which would enable a lower  $P_G$  to be able to obstruct fluid flow through a source channel with a higher  $P_S$ .

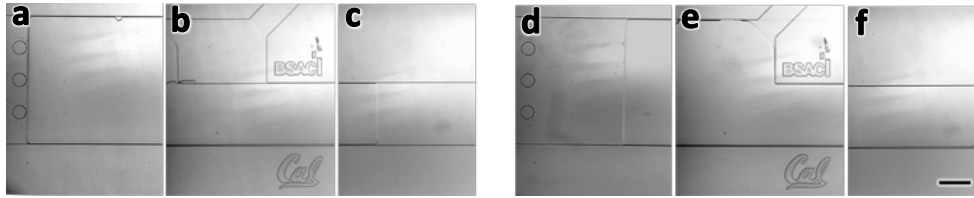
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### REFERENCES:

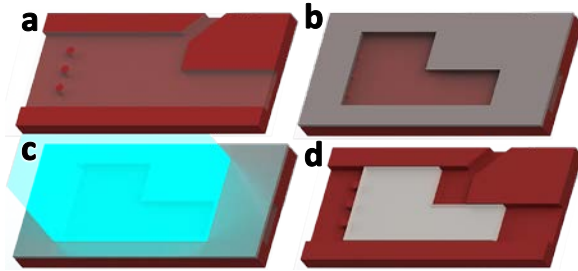
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- [2] B. Mosadegh, *et al.*, Lab Chip (2011)
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- [4] R. D. Sochol, *et al.*, MEMS 2013 (Accepted)



**Figure 1** Illustrations of the pressure gain device concept. (a) The device consists of a piston which slides within a microchannel with three fluidic terminals: a high-pressure ‘source,’ a low-pressure ‘gate,’ and an atmospheric pressure ‘exhaust.’ (b) In the ‘open’ configuration, the high pressure from the source forces the piston to rest against microposts on the gate side. (c) When pressure at the gate reaches passes a threshold pressure, the piston slides from the gate to the source (i.e., against the pressure gradient). The exhaust terminal serves as a vent for fluid trapped by the piston. (d) The piston includes two different cross-sectional areas – the source area ( $A_S$ ) and the gate area ( $A_G$ ) – which are exposed to a source pressure ( $P_G$ ) and gate pressure ( $P_S$ ), respectively. This differential area ratio  $R$  allows force amplification, enabling a low gate pressure to control a higher source pressure, with static gain approximately equal to  $R$ .



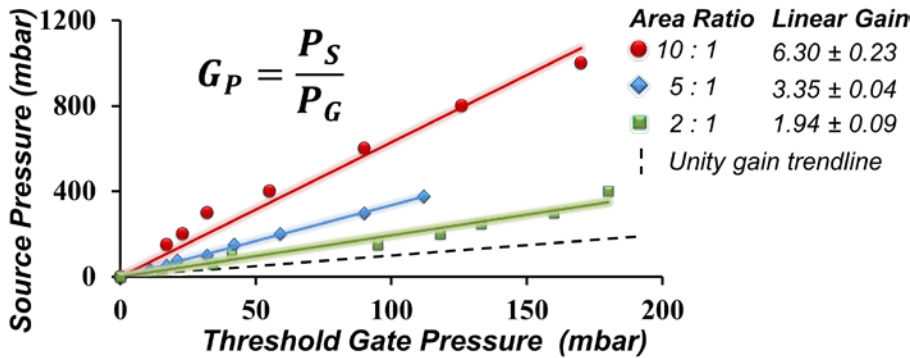
**Figure 2** Images of fabricated pressure gain device with an area ratio of 2:1 in open position (a-c) and closed position (d-e). Micrographs depict piston position in the gate, exhaust, and source regions of the device. Scale bar = 500  $\mu\text{m}$



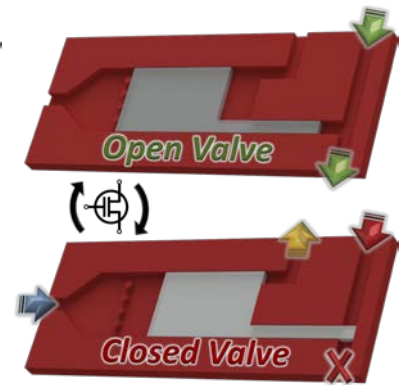
**Figure 3** The optofluidic fabrication process. (a) A photocurable liquid – PEGDA with 1% photoinitiator – is loaded into a PDMS microchannel. (b) A photomask is placed in contact with the device which is then (c) exposed to UV light, causing PEGDA to photopolymerize. The piston does not stick to the PDMS due to thin oxygen layers from the PDMS. (d) The mask is removed and the remaining photocurable liquid is drained from the device.



**Figure 4** COMSOL Multiphysics simulation the pressure gain device in the closed position. (a) Pressure field simulation. Gate pressure is set lower than source pressure, and exhaust is fixed at 1 ATM. (b) Fluid velocity field simulation. Source-exhaust flow can be minimized by decreasing gap size. The gap size is exaggerated for visualization purposes.



**Figure 5** Experimental measurements of threshold pressure  $P_G^*$  needed to move piston into ‘closed’ position, versus source pressure  $P_S$ . Measurements were taken for pressure gain devices with three separate area ratios. Gain was extrapolated from slope of the pressure curves. Gain = 1 line is included for reference.



**Figure 6** Conceptual illustration of pressure gain piston used as integrated valve mechanism in larger microfluidic device.